Proposal Kernels

Random Walk (MH)

$$\boldsymbol{\theta}'_{(i)} = \boldsymbol{\theta}_{(i-1)} + \boldsymbol{\epsilon},\ \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I})$$

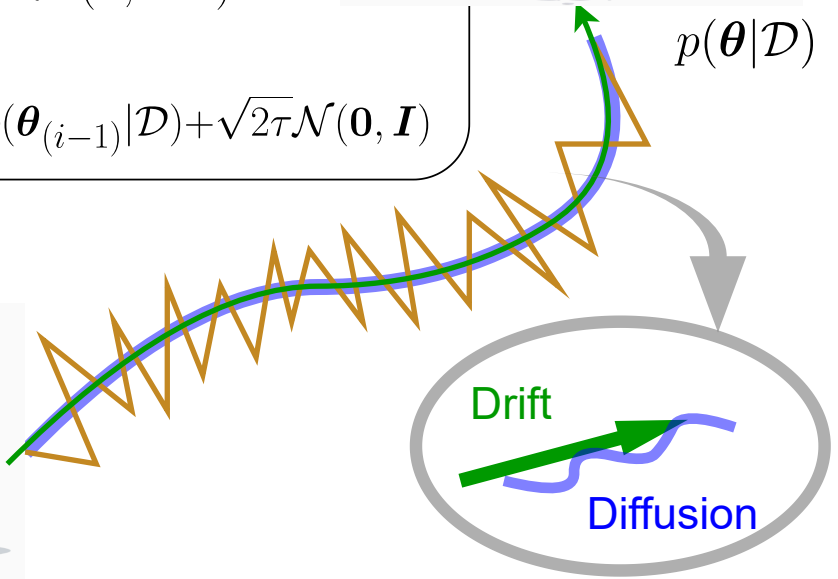
MALA

$$\boldsymbol{\theta}'_{(i)} = \boldsymbol{\theta}_{(i-1)} + \tau \nabla \log p(\boldsymbol{\theta}_{(i-1)}|\mathcal{D}) + \sqrt{2\tau}\mathcal{N}(\mathbf{0}, \boldsymbol{I})$$



$p(\boldsymbol{\theta}|\mathcal{D})$





Random Walk (MH)

MALA

$$d\boldsymbol{\theta} = \text{Drift} \cdot dt + \text{Diffusion} \cdot dW$$